

# SQL Assessment Test 2

## Questions and Expected Results

Keep in mind there is usually more than one way to answer these questions. For the solutions you can visit this link:

[https://docs.google.com/document/d/1swGZ0RG3KKqWqzmsl\\_qrMgjJ3lt39mtAJqRSMZy6Z-8/edit?usp=sharing](https://docs.google.com/document/d/1swGZ0RG3KKqWqzmsl_qrMgjJ3lt39mtAJqRSMZy6Z-8/edit?usp=sharing)

OR you can watch the solutions video walkthrough lecture.

These questions start off with the basics and then get continually more difficult.

1. How can you retrieve all the information from the cd.facilities table?
  - **Expected Result should look similar to this (with more rows):**

| Data Output |              |   |                         |  |            |  |           |  |               |  |                    | Explain | Messages | Notifications |
|-------------|--------------|---|-------------------------|--|------------|--|-----------|--|---------------|--|--------------------|---------|----------|---------------|
|             | facid        |   | name                    |  | membercost |  | guestcost |  | initialoutlay |  | monthlymaintenance |         |          |               |
|             | [PK] integer |   | character varying (100) |  | numeric    |  | numeric   |  | numeric       |  | numeric            |         |          |               |
| 1           |              | 0 | Tennis Court 1          |  | 5          |  | 25        |  | 10000         |  | 200                |         |          |               |
| 2           |              | 1 | Tennis Court 2          |  | 5          |  | 25        |  | 8000          |  | 200                |         |          |               |
| 3           |              | 2 | Badminton Court         |  | 0          |  | 15.5      |  | 4000          |  | 50                 |         |          |               |
| 4           |              | 3 | Table Tennis            |  | 0          |  | 5         |  | 320           |  | 10                 |         |          |               |

2. You want to print out a list of all of the facilities and their cost to members. How would you retrieve a list of only facility names and costs?

|   | Data Output                     | Explain               | Messages | Notifications |
|---|---------------------------------|-----------------------|----------|---------------|
|   | name<br>character varying (100) | membercost<br>numeric |          |               |
| 1 | Tennis Court 1                  | 5                     |          |               |
| 2 | Tennis Court 2                  | 5                     |          |               |
| 3 | Badminton Court                 | 0                     |          |               |
| 4 | Table Tennis                    | 0                     |          |               |

3. How can you produce a list of facilities that charge a fee to members?

- Expected Results should have just 5 rows:

| Data Output                                                                         |                                                                                                           | Explain                                                                                                             | Messages                                                                                                    | Notifications                                                                                              |                                                                                                                |                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | facid<br>[PK] integer  | name<br>character varying (100)  | membercost<br>numeric  | guestcost<br>numeric  | initialoutlay<br>numeric  | monthlymaintenance<br>numeric  |
| 1                                                                                   | 0                                                                                                         | Tennis Court 1                                                                                                      | 5                                                                                                           | 25                                                                                                         | 10000                                                                                                          | 200                                                                                                                 |
| 2                                                                                   | 1                                                                                                         | Tennis Court 2                                                                                                      | 5                                                                                                           | 25                                                                                                         | 8000                                                                                                           | 200                                                                                                                 |
| 3                                                                                   | 4                                                                                                         | Massage Room 1                                                                                                      | 35                                                                                                          | 80                                                                                                         | 4000                                                                                                           | 3000                                                                                                                |
| 4                                                                                   | 5                                                                                                         | Massage Room 2                                                                                                      | 35                                                                                                          | 80                                                                                                         | 4000                                                                                                           | 3000                                                                                                                |
| 5                                                                                   | 6                                                                                                         | Squash Court                                                                                                        | 3.5                                                                                                         | 17.5                                                                                                       | 5000                                                                                                           | 80                                                                                                                  |
|                                                                                     |                                                                                                           |                                                                                                                     |                                                                                                             |                                                                                                            |                                                                                                                |                                                                                                                     |

4. How can you produce a list of facilities that charge a fee to members, and that fee is less than 1/50th of the monthly maintenance cost?

Return the facid, facility name, member cost, and monthly maintenance of the facilities in question.

- Result is just two rows:

Data Output Explain Messages Notifications

|   | facid<br>[PK] integer | name<br>character varying (100) | membercost<br>numeric | monthlymaintenance<br>numeric |
|---|-----------------------|---------------------------------|-----------------------|-------------------------------|
| 1 | 4                     | Massage Room 1                  | 35                    | 3000                          |
| 2 | 5                     | Massage Room 2                  | 35                    | 3000                          |
|   |                       |                                 |                       |                               |

5. How can you produce a list of all facilities with the word 'Tennis' in their name?

- Expected Result is 3 rows

Data Output Explain Messages Notifications

|   | facid<br>[PK] integer | name<br>character varying (100) | membercost<br>numeric | guestcost<br>numeric | initialoutlay<br>numeric | monthlymaintenance<br>numeric |
|---|-----------------------|---------------------------------|-----------------------|----------------------|--------------------------|-------------------------------|
| 1 | 0                     | Tennis Court 1                  | 5                     | 25                   | 10000                    | 200                           |
| 2 | 1                     | Tennis Court 2                  | 5                     | 25                   | 8000                     | 200                           |
| 3 | 3                     | Table Tennis                    | 0                     | 5                    | 320                      | 10                            |
|   |                       |                                 |                       |                      |                          |                               |

6. How can you retrieve the details of facilities with ID 1 and 5? Try to do it without using the OR operator.

- Expected Result is 2 rows

Data Output Explain Messages Notifications

|   | facid<br>[PK] integer | name<br>character varying (100) | membercost<br>numeric | guestcost<br>numeric | initialoutlay<br>numeric | monthlymaintenance<br>numeric |
|---|-----------------------|---------------------------------|-----------------------|----------------------|--------------------------|-------------------------------|
| 1 | 1                     | Tennis Court 2                  | 5                     | 25                   | 8000                     | 200                           |
| 2 | 5                     | Massage Room 2                  | 35                    | 80                   | 4000                     | 3000                          |
|   |                       |                                 |                       |                      |                          |                               |

7. How can you produce a list of members who joined after the start of September 2012? Return the memid, surname, firstname, and joindate of the members in question.

- Expected Result is 10 rows (not all are shown below)

|   | memid<br>[PK] integer | surname<br>character varying (200) | firstname<br>character varying (200) | joindate<br>timestamp without time zone |
|---|-----------------------|------------------------------------|--------------------------------------|-----------------------------------------|
| 1 | 24                    | Sarwin                             | Ramnaresh                            | 2012-09-01 08:44:42                     |
| 2 | 26                    | Jones                              | Douglas                              | 2012-09-02 18:43:05                     |
| 3 | 27                    | Rumney                             | Henrietta                            | 2012-09-05 08:42:35                     |
| 4 | 28                    | Farrell                            | David                                | 2012-09-15 08:22:05                     |
| 5 | 29                    | Worthington-Smyth                  | Henry                                | 2012-09-17 12:27:15                     |
| 6 | 30                    | Purview                            | Millicent                            | 2012-09-18 19:04:01                     |
| 7 | 33                    | Tupperware                         | Hyacinth                             | 2012-09-18 19:32:05                     |
| 8 | 35                    | Hunt                               | John                                 | 2012-09-19 11:32:45                     |

8. How can you produce an ordered list of the first 10 surnames in the members table? The list must not contain duplicates.

- **Expected Result should be 10 rows if you include GUEST as a last name**

|    | surname<br>character varying (200) |
|----|------------------------------------|
| 1  | Bader                              |
| 2  | Baker                              |
| 3  | Boothe                             |
| 4  | Butters                            |
| 5  | Coplin                             |
| 6  | Crumpet                            |
| 7  | Dare                               |
| 8  | Farrell                            |
| 9  | Genting                            |
| 10 | GUEST                              |

9. You'd like to get the signup date of your last member. How can you retrieve this information?

- **Expected Result**
- **2012-09-26 18:08:45**

10. Produce a count of the number of facilities that have a cost to guests of 10 or more.

- **Expected Result**
- **6**

11. Produce a list of the total number of slots booked per facility in the month of September 2012. Produce an output table consisting of facility id and slots, sorted by the number of slots.

- **Expected Result is 9 rows**

|   | facid<br>integer |   | Total Slots<br>bigint |
|---|------------------|---|-----------------------|
| 1 |                  | 5 | 122                   |
| 2 |                  | 3 | 422                   |
| 3 |                  | 7 | 426                   |
| 4 |                  | 8 | 471                   |
| 5 |                  | 6 | 540                   |
| 6 |                  | 2 | 570                   |
| 7 |                  | 1 | 588                   |
| 8 |                  | 0 | 591                   |
| 9 |                  | 4 | 648                   |

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12. Produce a list of facilities with more than 1000 slots booked. Produce an output table consisting of facility id and total slots, sorted by facility id.

- **Expected Result is 5 rows**

|   | facid<br>integer |   | total_slots<br>bigint |
|---|------------------|---|-----------------------|
| 1 |                  | 0 | 1320                  |
| 2 |                  | 1 | 1278                  |
| 3 |                  | 2 | 1209                  |
| 4 |                  | 4 | 1404                  |
| 5 |                  | 6 | 1104                  |

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13. How can you produce a list of the start times for bookings for tennis courts, for the date '2012-09-21'? Return a list of start time and facility name pairings, ordered by the time.

- **Expected Result is 12 rows**

|    |  <b>start</b><br>timestamp without time zone |  <b>name</b><br>character varying (100) |  |
|----|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1  | 2012-09-21 08:00:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 2  | 2012-09-21 08:00:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |
| 3  | 2012-09-21 09:30:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 4  | 2012-09-21 10:00:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |
| 5  | 2012-09-21 11:30:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |
| 6  | 2012-09-21 12:00:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 7  | 2012-09-21 13:30:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 8  | 2012-09-21 14:00:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |
| 9  | 2012-09-21 15:30:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 10 | 2012-09-21 16:00:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |
| 11 | 2012-09-21 17:00:00                                                                                                           | Tennis Court 1                                                                                                           |                                                                                     |
| 12 | 2012-09-21 18:00:00                                                                                                           | Tennis Court 2                                                                                                           |                                                                                     |

14. How can you produce a list of the start times for bookings by members named 'David Farrell'?

- **Expected result is 34 rows of timestamps**